

# Designing AI-Mediated Digital Rituals through the Translation of Wabi-Sabi Aesthetics

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## Abstract

In contemporary life, digital communication technologies promise constant connectivity, while the accelerated pace of contemporary life has paradoxically increased interpersonal distance, making meaningful connections harder to maintain and causing many once-intimate relationships to fade. Drawing inspiration from the Japanese aesthetics of wabi-sabi and the tea ceremony principle of *ichigo ichie* (one time, one meeting), our study presents an online tea ceremony ritual prototype that integrates digital interaction, empathic dialogue design, and AI as a collaborative participant. Through preliminary evaluation, the study examines its potential impacts on emotional self-regulation, relationship repair, and cultural perception. The findings suggest that ritual processes characterized by rhythmic and sensory synchronization can facilitate emotional grounding and relationship building. Notably, introducing AI as a co-creative agent for poetic generation and memory reconstruction proved effective in shaping the experiential atmosphere of the interaction. This approach deepened conversations and elicited shared affective memories, offering new design directions for future AI-mediated digital ritual experiences.

## CCS Concepts

• Human-centered computing → Activity centered design.

## Keywords

Wabi-Sabi; AI-guided interaction; digital ritual design; practice-based research; remote social interaction; sensory synchronization

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## 1 Research Background and Motivation

Contemporary life has become increasingly fast-paced. With the widespread adoption of digital technologies and communication tools, the boundaries between work and personal life have gradually dissolved. Even during periods of rest, individuals often remain occupied with ongoing tasks, leaving limited space for personal reflection or emotional grounding. Without sufficient time to reflect on their own emotional experiences, individuals may find it increasingly difficult to sustain emotional presence in interpersonal relationships. As a result, maintaining close relationships becomes increasingly difficult, and formerly intimate connections may weaken through temporal distance and fragmented social interactions.

Within Japanese tea ceremony culture, the principle of *ichigo ichie*—literally translated as “one time, one meeting”—emphasizes the unrepeatable nature of each encounter and encourages participants to be fully present. Together with the aesthetic philosophy of wabi-sabi—which values imperfection, ambiguity, and sensory nuance—these cultural concepts frame interaction as a moment of attentiveness, slowness, and care[3]. In this way, everyday acts may come to be experienced as meaningful encounters, inviting participants to attune both to the present moment and to one another.

Responding to the erosion of interpersonal relationships, this research translates Japanese tea culture into a digital context. Building on prior HCI work that frames wabi-sabi as a design lens for slowness and reflection[2, 8], we extend this line of inquiry through an AI-mediated online tea ceremony. In this design, AI assists participants in assuming the role of a tea master, guiding the ritual flow and shared moments of attention to explore how emotional warmth and ritual depth may be sustained in remote social interaction.



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## 2 Research Objectives

First, this study aims to explore how the principles of *ichigo ichie* and the aesthetics of *wabi-sabi* can be translated into digital interaction design. Specifically, by reconfiguring ritual flows and reassembling essential elements of the tea ceremony, we seek to establish a framework for online social interaction that embodies ritual qualities and fosters emotional connection.

Second, it designs and enacts an online tea ceremony prototype characterized by ritualized structure and sensory synchronization. Drawing from traditional Japanese tea ceremony elements and sensory modalities—visual, auditory, and gustatory—we translate these components into a digital context and integrate them with interactive interfaces and AI-guided facilitation mechanisms. This approach formalizes the online tea ceremony experience while expanding its experiential richness.

Third, the research examines the effects of this interaction design on participants' emotional experiences, with particular attention to relational reconnection and perceptions of digital communication. This includes analyzing participants' emotional grounding, perceived intimacy, and evolving understandings of digital-mediated social interaction during and after the ritual experience.

Finally, it develops actionable emotional design strategies for digital social interaction. By focusing on process design and sensory synchronization, we explore how distributed participants may achieve shared affective states and moments of co-presence, offering design implications for future digitally mediated social experiences.

## 3 Related Work

With the widespread adoption of digital communication, establishing deep emotional connections in the absence of physical co-presence has become a critical concern within interaction design research. Lévy et al.[4] argue that ritual design differs fundamentally from everyday functional interactions, as it relies on structured processes and temporal orchestration to cultivate participants' attentiveness and appreciation of the present moment, thereby facilitating shifts in psychological states.

In their exploration of grief-related ritual design practices, Wakkary et al.[9] identify three key dimensions of embodied ritual experience: temporality, visibility, and emotional force. Their work demonstrates that guiding participants through multisensory design elements—such as sound, visual cues, and semantic framing—can effectively enhance ritual engagement and deepen users' emotional involvement.

Similarly, Garcia et al.[1] investigate emotional dynamics in online interaction and suggest that even in the absence of face-to-face contact, users' affective experiences are significantly shaped by interface design, interaction pacing, and the emotional tone of mediated content. These factors can contribute to emotional resonance and foster social connection within digitally mediated environments. Extending this notion of mediated connection to everyday practices, recent studies, such as the work by Panicker et al.[5] on mediated Human-Food Interaction, have explored the remote sharing of familiar rituals—like food and dining—through social interfaces. By embedding these everyday shared experiences into digital platforms, designers can further bridge the gap of physical

separation and enrich interpersonal connectedness. Complementing these perspectives, recent HCI research, such as studies by Kongot and Matz[2] and Tsaknaki and Fernaeus[8], has also drawn upon the Japanese aesthetic of *wabi-sabi* as a design resource, exploring how qualities of imperfection and transience can enrich digital materiality and user experience.

## 4 Method

This study adopts a practice-based research approach, exploring the potential application of *ichigo ichie* and *wabi-sabi* aesthetics within AI-generated and digitally mediated interaction contexts through the design and enactment of an online tea ceremony.

### 4.1 Activity Flow Design

Drawing on the canonical structure of Japanese tea ceremonies—such as entering the space, appreciating utensils, sharing tea, and expressing gratitude—this study translates and reconfigures ritual elements into a prototype suitable for remote digital interaction. The overall design integrates principles of sensory synchronization and ritualized interaction to support immersion and emotional connection in a digital environment. Ritual pacing and transitions between stages are facilitated by an AI tea master, with background human support assisting in generating contextual prompts and responses that guide participants into each phase of the ritual. The online tea ceremony consists of five sequential stages:

**(1) Preparatory Invitation:** An AI tea master generates poetic invitations and contextual information, serving as a ritualized prelude that introduces participants to the tea ceremony (Fig. 1). These help them transition from everyday communication into a more ritual-oriented interaction by establishing the atmosphere and intention of the ceremony.

**(2) Spatial and Emotional Grounding:** Participants are guided to prepare their physical surroundings and mental states. Ambient soundscapes are provided through shared audio playback during the online session, supporting the transition into a focused ritual mode.

**(3) Greeting and Reunion:** Initial greetings and emotional exchange are facilitated through AI-generated poems, memory phrases, and empathic topic cards.

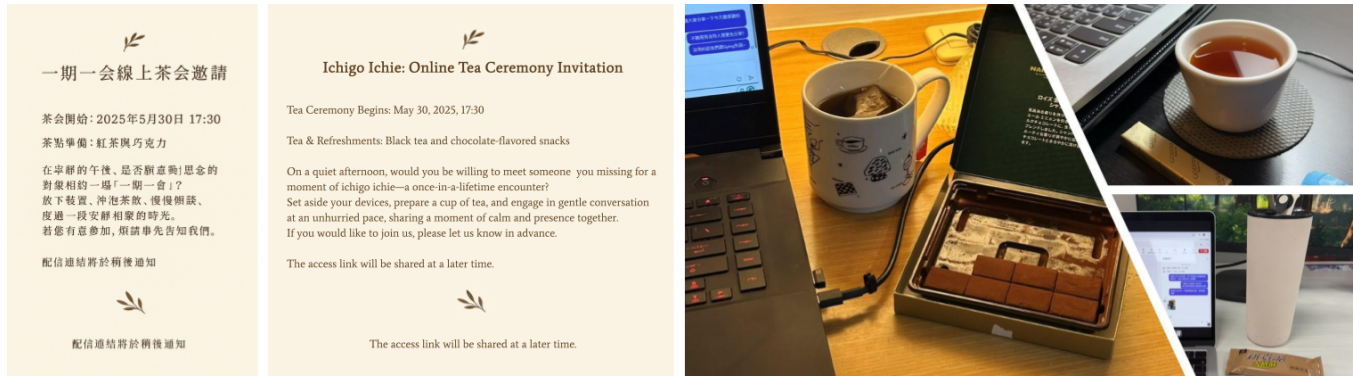
**(4) Synchronized Tea Tasting:** Participants simultaneously consume pre-arranged tea and snacks, fostering shared affect through synchronized gustatory experience across distance (Fig. 1).

**(5) Expression of Gratitude and Memory Preservation:** The ceremony concludes with expressions of gratitude, followed by an AI-generated personalized tea ceremony memory card, serving as a private digital keepsake that reflects the unrepeatable nature of *ichigo ichie* (Fig. 2).

### 4.2 Interaction Design and System Components

The interaction design aims to support emotional connection, temporal deceleration, and sensory attunement within digitally mediated contexts. To achieve these goals, the system comprises the following core interaction modules:

• **Poetic Generation Module:** The AI generates poetic lines in response to the tea ceremony context and participants' inputs,



**Figure 1: Left: Tea Ceremony Invitation Card, displaying the schedule, preparation instructions, and access information (original and English versions); Right: Tea and snacks arranged by participants according to the tea ceremony instructions.**

which are used as opening prompts or closing reflections to help establish a ritual atmosphere.

- **Soundscape Guidance Module:** Ambient natural soundscapes (e.g., forest, rain, wind) are employed to guide interaction rhythm and emotional pacing, helping participants experience the ceremony at a similar pace and feel present together despite physical distance.

- **Empathic Topic Cards:** Reflective prompts are introduced to encourage non-instrumental dialogue and facilitate emotionally resonant conversation between participants. These constitute the empathic dialogue design of the system, encouraging participants to share personal experiences, memories, and feelings.

- **Digital Memory Card Generation:** Elements from the tea ceremony, including selected food and beverages, conversational fragments, and AI-generated poetry, are synthesized into a personalized digital memory artifact.

Together, these modules constitute an AI-mediated interaction system designed to support ritualized remote social experiences.

## 5 Data Collection and Analysis

This study adopts a qualitative research approach, drawing on multiple data sources to holistically capture participants' interactions and experiential responses during the online tea ceremony. Participants (P1–P3) were recruited through the researchers' personal networks and were familiar with one another prior to the ceremony. Their existing interpersonal relationships made them suitable for exploring how digitally mediated rituals might support emotional connection. The collected data were analyzed through a preliminary thematic interpretation to explore initial insights. The following methods were employed:

- (1) **Participant Observation:** The researcher participated in the tea ceremony in the role of the tea master, observing and documenting participants' verbal interactions, non-verbal expressions, and patterns of interface use. This approach enabled close attention to participants' levels of engagement, sensory involvement, and transitions into ritualized states. The left part of Figure 2 shows the interface during the tea ceremony.

- (2) **Semi-Structured Interviews:** Post-ceremony interviews were conducted remotely and lasted approximately 15–30 minutes to explore participants' emotional trajectories, self-perceived experiences, evaluations of the interaction design, and evolving perceptions of interpersonal relationships. These interviews provided insight into how the ritual design influenced participants' emotional and relational experiences.

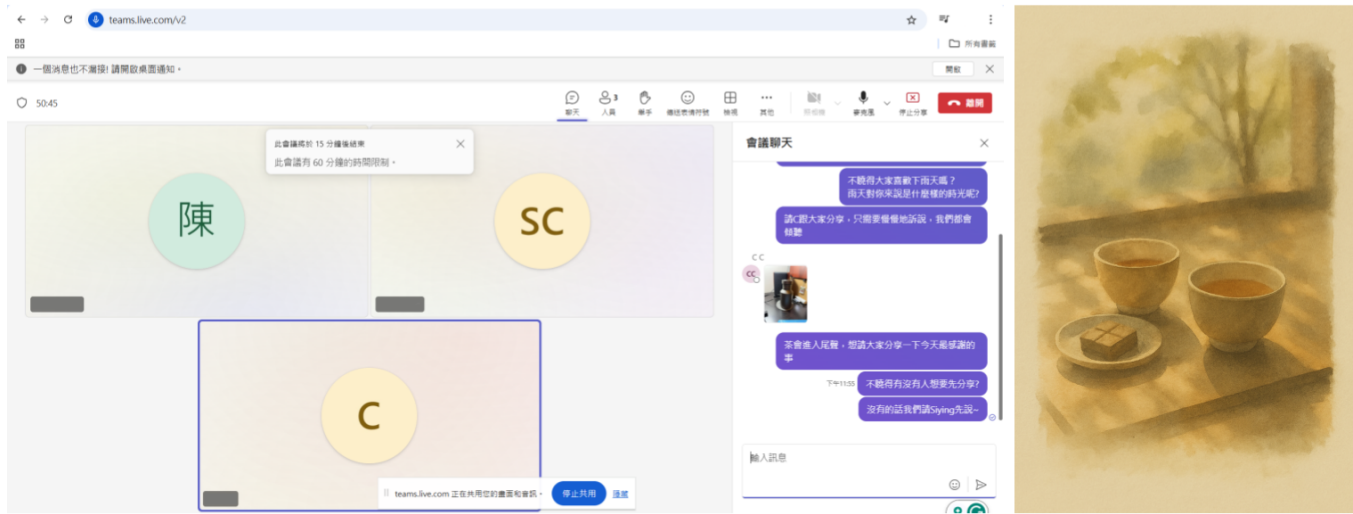
- (3) **Live Dialogue Recording and Transcription:** Conversational exchanges during the ceremony were recorded in real time. When appropriate, AI-assisted audio recording and automated transcription were used to generate verbatim transcripts, which served both as analytical material and as input for the generation of digital tea ceremony memory artifacts. All participants provided informed consent for recording and transcription, and identifying information was removed to protect participant privacy.

- (4) **Tea Ceremony Memory Card Generation:** The AI tea master synthesized elements such as tea selections, conversational fragments, poetic lines, and temporal impressions into personalized memory cards. These artifacts were privately shared between participants, embodying their memories of a unique and time-specific encounter inspired by the principle of ichigo ichie, while also serving as reflective traces of the interaction. The right part of Figure 2 shows the memory card, which allows participants to recall the weather, black tea, and chocolate shared during the tea ceremony.

## 6 Current Results

To date, this study has conducted one pilot test and one formal online tea ceremony, each lasting approximately 30 minutes and involving one human tea master collaborating with AI (P1) and two tea guests (P2 and P3). These initial enactments provide preliminary evidence for the feasibility of translating ichigo ichie and wabi-sabi principles into digitally mediated interaction design. As P3 reflected, *"The typos and imperfections in the AI-generated text felt unpredictable, and that made me think of wabi-sabi."*

The ceremonies were facilitated via Microsoft Teams and incorporated synchronized ritual pacing, sensorial guidance, and AI-generated poetic prompts. Participants described the AI tea master as a "gentle and poetic digital guide" that supported interaction rhythm and helped establish a ritual atmosphere. While



**Figure 2: Left: Use of Microsoft Teams in the Online Tea Ceremony. Beyond audio communication, the ceremony utilized interface features such as chat, and the platform was chosen for its support of audio playback from browser tabs; right: Tea Ceremony Memory Card Generation**

AI-generated language occasionally exhibited ambiguity or awkwardness (Fig. 3), participants interpreted these imperfections positively, sometimes relating them to wabi-sabi aesthetics and perceiving them as contributing to a sense of warmth and a more poetic experience in the human–AI interaction.

### 6.1 AI as a Ritual Co-Facilitator

In traditional tea ceremonies, the tea master draws on cultural knowledge and ritual expertise to guide participants through pacing, atmosphere, and emotional attunement. This study reimagines the tea master role within a human–AI co-facilitation model, in which ritual guidance is collaboratively constructed rather than solely human-led.

In this study, the AI tea master generates poetic prompts, empathic topics, soundscape suggestions, and a post-ceremony memory card. The human facilitator transforms these outputs into interaction guidance to support the ceremony, managing pacing, guiding transitions between stages, and providing technical support. Rather than replacing human agency, this division of labor positions AI as a scaffold for pacing, reflection, and helping establish a ritual atmosphere. As P2 noted, “*I realized I haven’t slowed down in a long time, and I haven’t sat quietly just to talk to someone.*” The design foregrounds a key question for human–AI interaction: to what extent can ritualized roles—traditionally grounded in human cultural authority—be meaningfully shared with AI in poetic, cultural, and affective domains.

Looking ahead, this work suggests the potential for more autonomous AI ritual facilitators that could support emotionally resonant remote interaction through structured yet flexible guidance.

### 6.2 Reflections and Insights: A Three-Level Analysis

#### Individual Level: Returning to Stillness and Inner Reflection

Participants experienced the tea ceremony as a moment of stillness that contrasted with everyday routines. As P3 described, “*it felt like a pause that I didn’t realize I needed.*” Synchronized tea tasting, soundscapes, and engagement with physical materials such as tea, cups, and snacks supported heightened sensory awareness and brief moments of emotional grounding and self-reflection.

#### Relational Level: Reopening Possibilities for Deep Dialogue

Shared preparation and synchronized interaction rhythms fostered a sense of closeness despite physical separation. As P2 reflected, “*I realized that a meaningful conversation doesn’t necessarily require meeting in person.*” Slower conversational pacing and expressions of gratitude shifted dialogue toward empathic presence and reaffirmed interpersonal bonds.

#### Cultural Level: Contemporary Expressions of Ritual Value

Participants reconceptualized ritual as an intentional structuring of attention rather than an elaborate form. The online tea ceremony provides preliminary indications that digitally mediated interactions can sustain emotional depth, challenging assumptions of online communication as inherently superficial.

## 7 Conclusion

### Communication Experiences Beyond Physical Distance

This study explored the potential of ritualized digital interaction to foster deep emotional connections and transcend physical distance. The enactment of the online tea ceremony suggests that even within remote digital environments, participants can experience a sense of co-presence, intimacy, and emotional warmth when interactions are guided by structured rhythms, sensory synchronization, and poetic contextual framing.

### Developmental Advantages of Human–AI Collaboration with an AI Tea Master

The collaborative role of the AI tea master in this study demonstrates distinctive potential and advantages. Beyond supporting the



**Figure 3: Text from the Tea Ceremony Invitation Card. The left part shows the extracted text, and the right part shows the corrected Traditional Chinese version.**

creation of a ritual atmosphere and maintaining interaction tempo, the AI facilitated deeper dialogue through poetic expressions and empathic topic prompts. This design enabled participants without specialized training to engage in a digital tea ceremony imbued with philosophical reflection and emotional depth. The findings suggest that AI tea masters may further evolve into independently guiding facilitators of digital rituals, extending the applicability of AI within interaction and experience design domains.

#### **Preliminary Convergence of AI Generation and Wabi-Sabi Aesthetics**

Although AI-generated text may exhibit structural or typographic irregularities, readers are often able to reconstruct meaning based on familiar linguistic patterns and contextual cues [6]. This phenomenon resonates with Jun'ichirō Tanizaki's reflections on imperfection and subtle ambiguity in *In Praise of Shadows* [7], which serve as a cultural reference for poetic sensibilities rooted in wabi-sabi aesthetics. In this context, the subtle deviations present in AI-generated language did not detract from the experience; instead, participants sometimes interpreted these irregularities as gentle imperfections that resonated with their understanding of wabi-sabi, allowing them to encounter moments of aesthetic resonance within digitally mediated communication.

## **8 Future Research Directions**

This study demonstrates the potential of integrating the tea ceremony with digital interaction design to support emotional connection. However, given the limited sample size, further research is needed to examine broader applicability and cross-cultural adaptation.

Future work will focus on three directions. First, interaction design elements such as empathic topic cards, soundscapes, and AI-generated poetry can be refined to enhance contextual sensitivity and emotional resonance across diverse cultural settings. Second, the role of AI may be extended from content generation toward co-facilitation, enabling AI to dynamically support pacing, reflection, and interaction flow in ritualized remote encounters. Finally, future research may explore AI systems that assume more autonomous ritual roles, investigating how structured ritual knowledge and contextual reasoning could inform emotionally grounded human–AI interaction.

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